



Consumer attention and market concentration in e-commerce: an agent-based perspective

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Abstract

This study explores how limited consumer attention influences market concentration in e-commerce. Consumer attention, a scarce resource amidst abundant product information, plays a crucial role in shaping market dynamics. Despite its importance, the effect of limited consumer attention on e-commerce market concentration has not been extensively studied. Using agent-based modeling, we examine the interplay of consumer behaviors, bounded rationality, and social interactions in complex markets. Our results reveal that e-commerce market concentration persists even without product differentiation among sellers. Notably, a negative correlation emerges between consumer attention and market concentration, consistent across different market sizes and social connection densities. These findings provide theoretical insights into market concentration patterns in the Internet economy and contribute to the broader understanding of market structure dynamics.

Keywords Limited attention · Market concentration · Social processing · Agent-based modeling · E-commerce

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1 Introduction

The development of e-commerce has made it easier to access product information compared to traditional business models. Online shopping eliminates geographical constraints and physical limitations for consumers, as they can simply swipe their fingers or click their mouse to navigate through the virtual world created by e-commerce platforms and browse through almost infinite product information, both product information from sellers and reviews from customers. However, due to the limited attention of consumers, the challenge they face is no longer a lack of information but rather information overload (Chen et al. 2009). This article aims to explore how consumers with limited attention make purchasing decisions in the context of information overload and how this translates into market outcomes at the macro-level.

Over the past decade, a trend of increasing market dominance by leading players has been observed across various industries. A handful of top sellers command a significant market share, overshadowing numerous smaller competitors who often remain unnoticed by consumers. For instance, the market shares of the top three bottled water brands in China are 26.5%, 21.3%, and 10.1%, respectively, collectively accounting for more than half of the bottled water market.¹ The market share of the top four sports shoe brands in China was 41.3% in 2010, increased to 61.4% in 2018, and reached 63.6% in 2021. This suggests that the intensity of oligopolistic competition among sports shoe brands is increasing, with the dominant players becoming more dominant and the less successful ones becoming less successful (Zhou 2021). Such phenomena can be quantified through market concentration, which refers to the extent to which a few firms dominate total sales, production, or capacity within a given market (Crouzet and Eberly 2019). From the perspective of industrial organization theory, most of China's markets have reached a serious oligopoly (Chen et al. 2009; Bai et al. 2014). Notably, such high market concentration ratios are not exclusive to China; they are also prevalent in countries with advanced commerce sectors, including Germany and the USA (Borsenberger 2015; Kwon et al. 2024).

While extensive research has been conducted to understand the factors driving market concentration, these studies predominantly focus on the seller's perspective, exploring themes such as market competition and product differentiation (Wei 2001; Dong et al. 2023), natural monopolies because of high infrastructural costs (Ducci 2020), industrial policies (Wei et al. 2023), and technological advancements by enterprises (Anderson et al. 2012; Dolata et al. 2018). These investigations suggest that a seller's intrinsic resources and market strategies significantly contribute to their market dominance. While these findings align with several managerial practices, they do not entirely account for all instances of market concentration. For instance, the wide disparity in sales among sellers of identical products on e-commerce platforms indicates that more than just product variations are at play.

¹ <https://www.credenceresearch.com/report/china-bottled-water-market>.

In this study, we propose that another crucial player in the market—consumers—should not be ignored when explaining the phenomenon of market concentration, particularly considering the role of limited consumer attention. A significant challenge in the Internet economy is the issue of information overload, which has become increasingly prevalent (Lee and Lee 2004; Chernev et al. 2015; Wan et al. 2009). Decision-making difficulties are no longer caused by a lack of information but by an overload of it. In this context, consumer attention has emerged as a pivotal factor, and competition for attention has become a strategic choice that companies cannot afford to overlook (Evans 2020; Ding et al. 2020). Furthermore, limited attention has a significant impact on the way consumers process information and make decisions. When consumers are unable to focus on all available information, they tend to rely on heuristics or mental shortcuts to make choices (Tversky and Kahneman 1974). These methods do not necessarily lead to optimal choices; instead, they result in satisfactory decisions influenced by factors like peer recommendations or sales rankings, based on a limited scope of search (Hervas-Drane 2015). Social influence, therefore, becomes a crucial factor, as individuals refer to the actions, opinions, and endorsements of peers to help simplify their decision-making process.

Although the concept of limited attention has been previously introduced and developed (Goldhaber 1997; Frank 1999; Lanham 2006; Evans 2020), the direct correlation between consumer attention and market structures remains insufficiently addressed. For example, previous studies, such as Huotari (2017), examine the impact of interactions between diverse market participants—namely consumers and differentiated firms—on platform performance, highlighting the reciprocal influence of firm strategies and consumer preferences. Similarly, Fatas-Villafranca et al. (2019) simulate co-evolutionary dynamics between consumers and profit-maximizing firms, underscoring the role of firm heterogeneity in shaping outcomes. In contrast, our study isolates the role of consumers in shaping market concentration. By assuming firm homogeneity in product offerings, we specifically examine how consumer-side factors, especially limited attention, impact market outcomes. Yanagita and Tamotsu (2008, 2010) relax the assumption of consumer rationality in their model, incorporating a larger pool of market participants. Through numerical simulations, they underscore the impact of sellers' strategies and consumers' bounded rationality on market structures, particularly oligopolies. Their contribution is valuable in linking bounded rationality to market structure. However, their model adheres to the traditional economic view that sellers primarily drive market concentration, without clearly explaining the extent to which consumers' bounded rationality affects this process.

Viewing the market as a dynamic complex system, composed of numerous interacting agents exhibiting limited attention on purchase, this study seeks to analyze the evolution of market concentration as an aggregate result of the decentralized decision-making of consumers who are constrained by limited attention. Our research assumes that consumers cannot fully comprehend all sellers and may be unable to discern the optimal option due to limited cognitive resources. Consequently, consumers may seek recommendations from their peers, and therefore social influence plays a role. On the other hand, we control for variations in sellers' behaviors to emphasize a theoretical possibility that the market concentration can be high even in

a market with identical products. We intend to show that even without objective differences between products, consumer feedback can “create” subjective differences among objectively identical products, conferring a competitive advantage on some sellers. This advantage can be amplified in future transactions by influencing potential consumers’ choices and in the extreme case could lead to a “winner-takes-all” outcome. In our view, this is an important aspect of understanding concentration in e-commerce markets.

Social simulation plays an important role in systematically studying emergent social dynamics. One of the most prominent methodologies used in social dynamics is agent-based modeling (ABM) (Epstein and Axtell 1996; Macy and Willer 2002). The importance of ABM in this study is highlighted through a comparison with traditional methods. For example, traditional game theory is a formal modeling approach used to examine individual behaviors and their outcomes (Osborne and Rubinstein 1994). However, it is not an ideal research method for our study because it is more focused on equilibrium states and strategic outcomes rather than dynamics (De Marchi and Page 2014). This may not capture the continuous and nuanced changes in social systems as effectively as ABM, which allows for complex systemic outcomes beyond static equilibria or simple patterns. Moreover, ABM facilitates computer-based experimentation, allowing for a systematic evaluation of how specific theoretical assumptions and contextual factors shape complex social dynamics (Gilbert and Troitzsch 2005). Another popular approach involves empirical methods, such as statistics and surveys. While they are instrumental in exploring factors influencing consumer decision-making, they often fall short of capturing the dynamic and psychological processes that form the basis of these decisions. The aim of our research lies not just in identifying the statistical relationship between variables but rather in elucidating how attention theoretically operates and influences market concentration, thereby complementing empirical research. ABM’s capacity to simulate various theoretical scenarios provides a powerful means to enhance our understanding and interpretation of empirical data (Li and Jager 2023). Moreover, social simulations are based on computational implementations of causal mechanisms and as such allow for the examination of whether specific causal mechanisms can reproduce the empirically observed phenomena, thus contributing to theory development.

The study may deepen our comprehension of how consumer decision-making influences market dynamics in e-commerce by merging insights from consumer attention literature with ABM. It focuses on exploring the complexity of consumer behavior when purchasing undifferentiated products online. Through ABM simulations, the research aims to clarify theoretically how individual consumer attention shapes market concentration dynamics from a collective perspective. Additionally, by examining the interplay between consumer attention and market concentration, the study holds the potential to provide valuable insights for policymakers, businesses, and researchers navigating the rapidly evolving e-commerce landscape.

The remainder of the paper is organized as follows: Sect. 2 presents the theoretical background. Section 3 outlines the model’s specifics. Section 4 details the simulation experiment design and model outcomes. Section 5 analyzes the findings and compares them with prior research. Conclusions and practical implications are presented in Sect. 6.

2 Theoretical background

2.1 Market concentration ratio

Market concentration refers to the extent to which a relatively small number of firms dominate a particular market or industry. It is typically measured using metrics such as the market concentration ratio (Schwartzman 1960; Jadow 1981; Mazzucato 2000; Bykova 2017), which assesses the proportion of economic indicators, such as profits or sales in a given market, attributed to these firms. This ratio is a critical measure for evaluating market competition and dynamics. A high market concentration ratio means a small number of firms hold a large share, while a low ratio indicates the market is more evenly distributed among many competitors. This concept differs from that of an industry cluster, which refers to a geographical phenomenon of the high density of enterprises related to a specific industry within a particular area (Krugman 1991; Feser and Edward 2000). The market concentration emphasizes the concentration of companies selling similar products, reflecting the relative scale of sellers, regardless of their geographic distribution and relationship between upstream and downstream enterprises.

The concept of market concentration within e-commerce differs markedly from traditional interpretations. While stability in concentration in conventional markets often arises from factors like product quality and pricing, the digital realm introduces additional dynamics. The e-commerce setting is marked by potentially fluctuating levels of concentration, contrasting with the steadiness observed in traditional markets. For example, during Alibaba's "Singles' Day" shopping festival (Wu et al. 2016), certain products or brands may receive more attention, leading to temporary spikes in market concentration. However, as these promotional events conclude and consumer attention shifts, market concentration may rapidly decline, with other products or brands gaining prominence in subsequent days or weeks. This fluctuation in market concentration reflects the dynamic nature of the e-commerce landscape, where consumer behavior and attention play significant roles in shaping market dynamics.

2.2 Limited attention

The significance of consumer attention has grown substantially in the study of consumer behavior, given its crucial role in the decision-making process (Vriens et al. 2020; Peschel et al. 2019). The concept of attention in psychology encompasses a range of mechanisms, including alertness, orientation, and selective attention processes like detection, facilitation, and inhibition (Schmidt 2001; Tomlin and Villa 1994). Human attention is limited due to the structure of the human brain, specifically the limited capacity of neurons for memory storage. This inherent restriction results in attention being restricted over a given period. Limited consumer attention refers to the finite capacity of individuals to focus on and consider a limited number of options or brands within a particular market at any given time. For instance, when a consumer is in the market for a laptop, they may evaluate several brands

such as Apple, Dell, HP, and Lenovo. However, due to the constraints of their attention, they are unlikely to consider every available brand, which implies that, despite the availability of many options, consumers can only consider a subset of brands or products during the decision-making process. This limitation is echoed in the concept of bounded rationality in economics and management, as proposed by Simon (1955) and Kahneman (1979). These scholars argue that constraints on information processing capacities prevent market participants from making entirely rational decisions, as posited in neoclassical economics. Instead, individuals often make decisions that are merely satisfactory. Furthermore, Simon proposed the concept of procedural rationality as a complement to outcome rationality. This encompasses the investment of cognitive effort in the optimization process. It is not rational to spend excessive effort on many decisions.

While the exact causes of limited rationality are not fully understood, limited attention is acknowledged as a significant factor. The rise of the information age in the late 1990s highlighted the scarcity of attention resources, sparking widespread research interest. Goldhaber (1997), often regarded as the pioneer of attention economics, marked the transition to an era dominated by attention scarcity amid information abundance. Wang (2000) developed economic models to analyze attention allocation efficiency. Drawing on China's experience, Zhang (2001) formulated concepts within the attention economy. Chernev (2003) demonstrated through behavioral experiments how consumer choices could be influenced by the number of available options. Moreover, studies by Huang et al. (2009) and De Los Santos et al. (2012) indicate that consumer online searches are typically limited to a few sellers.

Drawing from existing literature and our observations of consumer purchasing patterns, this article conceptualizes consumer attention as the search capacity preceding a purchase. Specifically, it refers to the number of sellers a consumer explores and considers in the market within a given time frame. This concept is similar to the consumer's consideration set, which measures the breadth of product options contemplated during the purchasing process (Shocker et al. 1991; Gu et al. 2012). We acknowledge that the scope of limited attention encompasses the entire consumption process, potentially influencing not just the search phase but also the later stages of decision-making. While it might appear reductive to equate limited attention solely with restricted search activity, this definition is adopted in our study to maintain focus on the primary research questions and to simplify the discussion. This approach allows us to examine the concept of limited attention within a manageable framework for this research.

2.3 Social processing

In social psychology, social processing is understood as a means by which individuals learn about unfamiliar behaviors through observing the actions of others (Jansen and Jager 1999). This cognitive strategy is employed to reduce effort, fitting into a broader framework of optimization. Social processing often involves individuals engaging in comparative analyses with other consumers through social interactions. Introduced by psychologist Leon Festinger in 1954, the concept of social

comparison posits that people look to the behavior of similar others when unsure of how to act. Additionally, the theory of reasoned action (Fishbein and Ajzen 1977) suggests that the behaviors of others are often viewed as normative.

In our study, we focus on consumers with limited attention who are likely to seek knowledge from others to make informed purchasing decisions. Social influence operates through the connections between individuals, closely tied to the social network that serves as a channel for interaction, information dissemination, and opinion formation (Festinger et al. 1954). Following the majority is often perceived as a safe choice and is referred to as normative influence by Cialdini and Goldstein (2004). This concept describes the social pressure individuals face to conform to societal norms while also leveraging the collective wisdom of the majority. For example, a product recommended by a majority of friends may be perceived as a reliable option in the absence of personal knowledge. This is particularly evident in the context of word of mouth (WoM). WoM involves the verbal sharing of information and experiences about a product (Schindler and Bickart 2005). It encompasses individuals conveying their experiences or insights, and expressing personal preferences regarding a specific product with others in their social or professional networks. Modern-day examples include online reviews and testimonials. Consumers frequently consult these reviews, and positive feedback can lead to a favorable WoM impact. This can be seen as a continuous social learning process (Axelrod 1981), where consumers gather information and modify their purchasing decisions based on others' experiences.

Many studies have found that higher-quality options (whether products, ideas, candidates, etc.) are not necessarily the most successful. For example, Salganik et al. (2006) discovered through simulated experiments that the success of cultural products is only partially related to quality factors: generally speaking, high-quality products do not perform poorly, and low-quality products do not perform very well, but other outcomes are possible. Social influence is a key factor. Later consumers are influenced by the market share achieved by earlier consumers, leading to a Matthew effect where products that initially gain an advantage due to random factors continue to expand their advantage.

However, other studies have pointed out that besides the above-mentioned “self-reinforcing” mechanism, the market also has a “self-correcting” mechanism. According to this view, “gold will shine” eventually, and although high-quality products may be temporarily disadvantaged due to factors such as social influence, they will ultimately prevail and become the choice of the majority (van de Rijt 2022).

We argue that while the self-correcting mechanism indeed plays an important role as an inherent force in the traditional market economy, its effectiveness is conditional. As van de Rijt (2022) noted, this mechanism requires some subsequent consumers to make independent judgments and autonomous choices, disregarding social influence, which is not always feasible. A classic example is the PC keyboard layout. The QWERTY layout may not be the most efficient, but because users have become accustomed to it, alternative layouts cannot survive. The key argument of our paper is that in the e-commerce retail market, the limited attention of consumers significantly strengthens the cumulative advantage. The more limited the consumers' attention, the more their purchasing decisions

rely on recommendations from platforms and friends, making the impact of social influence more significant (note here that social influence is more of a help to individuals with limited rationality), reducing the space for autonomous choice.

This study aims to delve into the dynamics of consumer attention and its impact on market concentration within e-commerce realms. With a focus on consumer decision-making processes, particularly the role of individual attention and social relations, it endeavors to shed light on their impact on market dynamics. The study revolves around three core entities: consumers, sellers, and the market itself. The research objectives are twofold: first, to examine the impact of individual attention and social relations on consumer behavior, and second, to investigate how the collective dynamics of market concentration are influenced by individual consumer attention and evaluations of products or sellers. To achieve these objectives, the study integrates three foundational theories into its framework (illustrated in Fig. 1): limited attention theory clarifies the role of consumer attention in shaping decision-making, social processing theory investigates the impact of social interactions on consumer behavior, and market concentration acts as a metric to evaluate market structure.

Employing ABM, the study develops a computational model to simulate consumer behavior and interactions within e-commerce settings. Through experimentation within the computational system, diverse scenarios are simulated to observe emergent market dynamics. Subsequently, simulation data are collected and analyzed to uncover patterns, trends, and correlations between consumer attention, market concentration, and other relevant variables.

In summary, this research framework provides a systematic methodology for exploring the interplay between consumer attention, decision-making processes, and market concentration in e-commerce environments. By utilizing ABM and computational modeling techniques, the study aims to offer valuable insights for understanding and managing market dynamics in the digital marketplace.

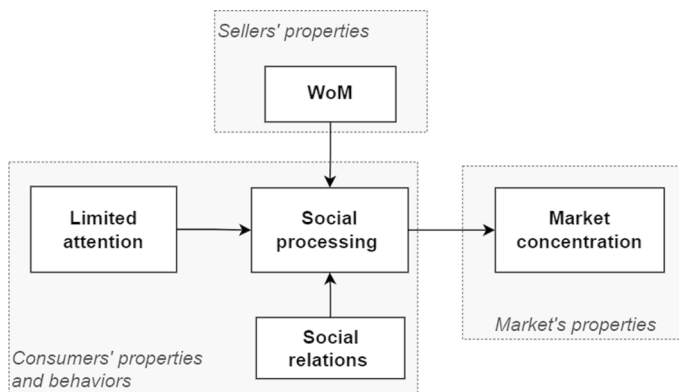


Fig. 1 Schematics of the conceptual framework

3 The model

In this research, we developed an agent-based model using the NetLogo simulation environment (Wilensky 1999). The primary aim of this model is to examine the relationship between limited consumer attention and market concentration. The model includes several key components: consumers, sellers, bounded rationality, networks, and social interactions. It focuses on how consumer purchase decisions evolve, influenced by product information and feedback from peers, and guided by specific behavioral rules.

The use of an agent-based approach and these comprehensive components is central to our research. We intend to conduct simulation experiments to reveal the theoretical relationship between consumer attention constraints and market concentration. These experiments are designed to show how the overall market structure emerges from these individual interactions.

3.1 Agent properties

In our consumer purchasing decision model, we define two main types of agents: sellers and consumers. The sellers are characterized by four main attributes: their total number, the quality of their products, sales performance, and the influence they exert through word of mouth. Consumers, on the other hand, are characterized by three primary attributes: their total number, the extent of their attention span, and their social networks.

3.1.1 Sellers

In our model, a seller refers to any individual or entity offering products, services, or financial assets for sale (Walter et al. 2001). In this research, we assume that sellers can produce an endless supply of identical products. Thus, choosing a seller in our model is analogous to selecting a product. Each seller in the model has distinct attributes and characteristics.

Number of sellers. We assume the presence of N_s sellers in the market, represented as $S = \{S_j | j = 1, 2, \dots, N_s\}$, where each S_j is a unique seller.

Quality of products. Product quality is a multifaceted concept that includes durability, reliability, user-friendliness, and repairability, among other factors (Mussa and Rosen 1978). Product quality is a critical determinant in consumer purchasing behavior. It is important to emphasize that, based on the research objectives of this paper—specifically, the possibility of market concentration in a homogenous product market influenced by limited consumer attention and social influence—all product qualities in the model are assumed to be identical. This assumption is made to avoid any objective differences between the products.

Sales of sellers. Sales refer to the quantity of products promoted and sold by a company within a specific period (Jobber and Lancaster 2006). In our study, this concept translates to how often consumers choose a particular seller to buy products.

Sales of sellers are denoted as $SS = \{SS_j | j = 1, 2, \dots, N_s\}$, where each SS_j represents the sales of seller S_j , indicating the frequency of consumer selection for that seller within a certain time frame.

Word of mouth (WoM). WoM is used to denote the collective evaluation of a product or seller by consumers. WoM is based on the aggregated assessments of interconnected consumers through their social networks regarding a specific seller, which is a critical mechanism in the diffusion process, representing informal, interpersonal communication (Van Eck et al. 2011).

In our model, similar to Van Eck's approach, WoM is simulated by allowing agents to share purchasing information with their network neighbors, impacting their likelihood of adoption. WoM functions as a metric to identify the most highly regarded seller among the neighbors within their consideration set. Essentially, the seller most frequently considered by the local majority is deemed the best seller with the highest WoM. Specifically, if a consumer has 5 network neighbors, each with a consideration set size of 3 (meaning each neighbor considers 3 candidate sellers), the consumer will evaluate a total of 15 candidate decisions. Subsequently, these options are ranked based on their frequency of occurrence among the neighbors. The option with the highest frequency is regarded as the most popular choice among the neighbors, thereby indicating the seller with the highest WoM. Importantly, this WoM is not a global variable but rather a subjective evaluation unique to each consumer, reflecting their individual perceptions and preferences. This setting intentionally bypasses the actual process by which consumers make evaluations within a rating system. Instead, it simplifies the assessment to a statistical aggregation of the majority choice within the consumers' consideration sets. Therefore, the WoM in this model is akin to the normative influence described by Cialdini and Goldstein (2004).

3.1.2 Consumers

Consumers are key agents in the market, participating in the purchase of products or services. They exhibit various attributes and possess decision-making capabilities, with their choices influenced by inherent properties, behavioral patterns, and interactions among themselves.

Number of consumers. The market is typically composed of numerous consumers. We denote the total number of consumers as N_c , represented as $C = \{C_i | i = 1, 2, \dots, N_c\}$, where each C_i represents an individual consumer (as expressed in Eq. 1).

$$C_i = \{\text{properties, behaviors}\}, i = 1, 2, \dots, N_c \quad (1)$$

The behaviors of consumer C_i particularly relate to their actions in the purchasing process. We will discuss the mechanisms behind these behaviors in Sect. 3.3.

Consumer's attention. Previous studies have shown that consumers display bounded rationality, incomplete information, and cognitive limitations (Simon 1955; Wittek et al. 2013). We define these attributes collectively as "consumer's attention" (CA), which is consistent across all consumers. CA is quantified as the pre-purchase search

capacity, specifically the number of sellers a consumer explores in the market over a given period. CA is a natural number greater than 1, but cannot exceed N_s . A higher CA means the consumer can consider more products before purchasing, while a lower CA indicates a more limited range of product considerations.

To illustrate this, we introduce the concept of a “consideration set” (CS) (as defined in Eq. 2), which reflects a consumer’s attention level. The size of the CS is directly proportional to the consumer’s attention.

$$CS_i = \{s_j, \dots s_k, \dots s_l\} \quad (2)$$

Here, CS_i is the consideration set for consumer C_i , and s_j, s_k, s_l are elements within the set of sellers S . The number of items in the CS is tied to consumer C_i ’s attention level (CA). For example, if CA is 3, then there will be 3 items in the CS_i .

Social network. Decision-making among consumers is a social activity, influenced by peers, whether in close networks or broader social circles (Li and Jager 2023). Consumers’ choices are significantly affected by their social interactions, where they gain insights about sellers. This social influence is closely connected with the dynamics of their social networks. The next section will explore the communication networks that encompass the consumer population.

3.2 Consumer’s communication network

In this study, we conceptualize the interplay of consumers within the market as constituting a social network. The configuration of this consumer social network adheres to the algorithm applied in earlier ABM studies, for example, by Li and Jager (2023). According to this algorithm, individuals exhibit a greater propensity to establish connections with those in their immediate vicinity as opposed to those situated at a distance. This proximity can be defined in terms of both geographic and relational closeness. Initially, consumers are distributed randomly across the spatial domain, and subsequently, they form undirected connections with nearby individuals. These interconnected consumers within the market can be likened to a group of acquaintances who possess the capacity to exert normative influence upon one another.

In this study, we employ the network connectivity coefficient (NCC) as a metric for delineating the connectivity status of the social network encompassing consumers throughout the entire market (Wang et al. 2018). NCC is specifically defined as the ratio of the present connections number (PCN) to the total number of connections that would be realized in the case of complete connectivity, which corresponds to the total connections number (TCN) within the network. NCC is a metric used to assess the density of a network. A higher NCC indicates a more dense network, while a lower value indicates a less dense network. This calculation can be performed as detailed in Eqs. 3 and 4.

$$NCC = \frac{PCN}{TCN} \quad (3)$$

$$\text{TCN} = C_{N_c}^2 = \frac{N_c(N_c - 1)}{2} \quad (4)$$

where N_c signifies the number of consumers. The network connectivity coefficient (NCC) falls within the range $[0, 1]$. An NCC value of 0 indicates a complete absence of communication among consumers. Values in the range $0 < \text{NCC} < 1$ denote the presence of communication within the market, with higher NCC values implying a greater extent of consumer interaction. An NCC of 1 signifies that the social network encompassing all consumers in the market is fully connected, allowing for communication between any pair of consumers. Figure 2 illustrates examples of market networks comprising 20 consumers, each featuring distinct NCC values.

3.3 Consumer's decision-making process

Consumer perceptions of sellers are shaped not just by their individual buying experiences but also by the influence of word of mouth within interconnected social networks, such as friends or neighbors. The process of consumer purchasing decision-making follows a series of steps, as illustrated in Fig. 3. The flowchart illustrates consumer decision-making across three interconnected levels: the collective level, where market-wide dynamics and concentration are evaluated; the individual level, focusing on consumers' consideration set formation and purchasing decisions; and the social network level, which incorporates word of mouth and social influence from neighbors in updating the consideration set.

1. *Search*: When confronted with a novel market, consumers embark on their decision-making journey by making an initial selection from a myriad of potential sellers. Initially, consumers will randomly explore a pool of CA potential sellers with equal probability from all available sellers to create their initial consideration set.
2. *Inquire*: During this phase, consumers seek information about sellers from their interconnected peers. Here, "word of mouth" encapsulates the wisdom shared among fellow consumers. Each consumer maintains their own consideration set, which is shaped by the following rules:

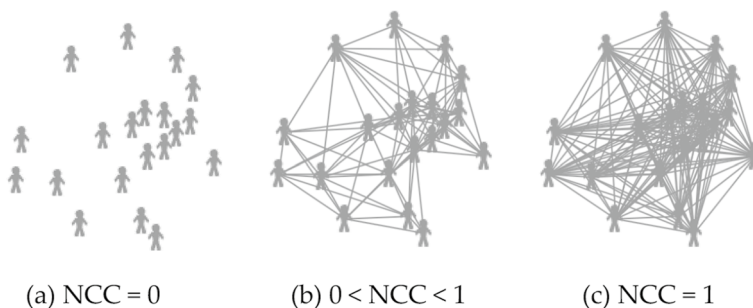


Fig. 2 Social network connections status of consumers with different NCCs

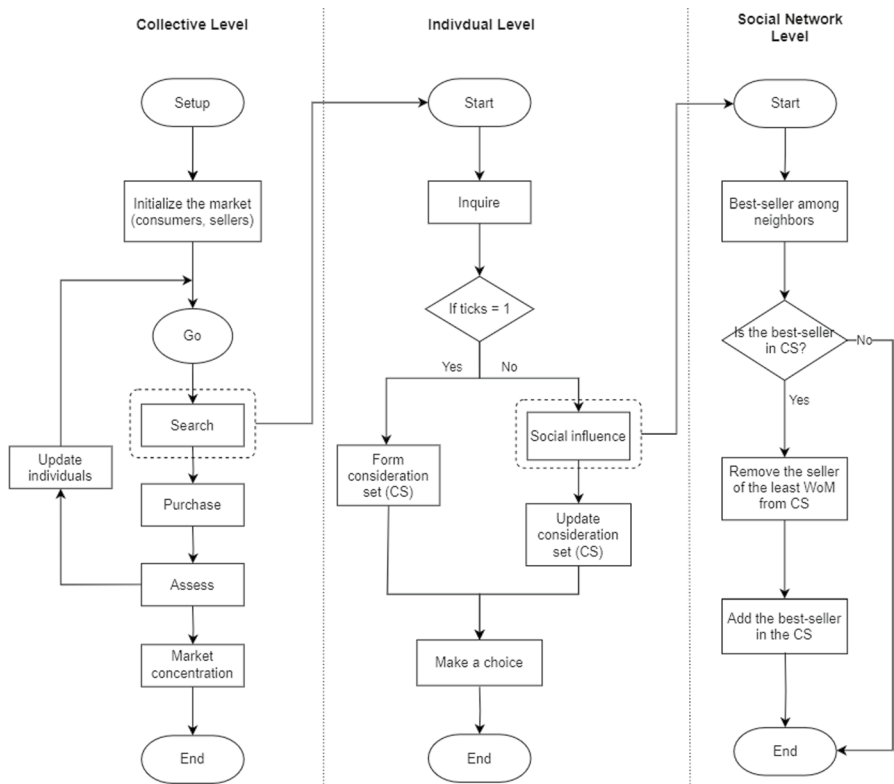


Fig. 3 Behavior mechanism of consumer's purchasing decision-making

- (a) The first item in the consideration set is either the seller they previously chose (if there is prior purchasing experience) or a randomly selected seller (in the case of a completely new market).
 - (b) The second item is the most considered seller by their interconnected peers, or a random choice if the market is entirely new.
 - (c) The remaining (CA-2) items are randomly chosen to complete the consideration set (CS), ensuring that the total number of items equals CA. The primary objective of these first two steps is to form each consumer's consideration set.
3. *Purchase*: It is posited that consumers exhibit no distinctive preferences among items within the consideration set (CS) in this research. Specifically, we posit a scenario wherein the importance of each item within the CS is deemed uniform. Consequently, during the process of making a purchasing decision, consumers engage in a random selection process, wherein a single item (representing a particular seller) is chosen at random from the CS, composed of CA items as determined in the previous stage.

4. *Assess*: Following a purchase, consumers update their consideration sets by incorporating their last buying experience and the preferences of their peers. Specifically, the product they just bought is added as the first item in their consideration set. Additionally, they identify the most frequently selected product from their neighbors' consideration sets and include it as the second item. Concurrently, the sales volume for the selected seller increases by the number of consumers who make a purchase. After updating their consideration sets, consumers establish connections with the new sellers whose products have entered their consideration sets and disconnect from sellers whose products are no longer included.

A time step represents one complete cycle of exchanges between buyers and sellers. After each time step, the market calculates each seller's sales, ranks all sellers, and determines concentration levels. The model then moves to the next time step, starting a new round of trading and repeating the decision-making process. Such settings reflect a pattern of repeated purchases; for example, toiletries are used daily and require regular replacement.

3.4 Market concentration

The dynamic interaction between consumers and sellers gives rise to the formation of a market at the macro-level. The pivotal element characterizing the market's essence is its degree of concentration, which quantifies the market share held by the dominant sellers. The positions of these sellers in the sales hierarchy tend to shift over time. While some sellers might experience a surge in sales, others could contend with extended periods of limited or no sales and, in some cases, eventual elimination, depending on shifting consumer preferences within each time frame.

To assess this market structure, a concentration ratio (CR) is employed, which sums up the percentage market shares of the largest sellers within a given market (as expressed in Eq. 5). Specifically, an n -seller concentration ratio (CR $_n$) serves as a widely used metric to gauge the market's composition, reflecting the combined market share of the n largest sellers operating within it. A higher CR $_n$ signifies a more pronounced concentration within the market. The choice of the value of n depends on several factors, such as the industry's context, the number of enterprises in play, and the research objectives. Typically, research studies opt for values like 3, 4, or 8, as indicated in previous works (Tirole 1988; Wei 2002). For the purposes of this study, we employ CR3, representing the proportion of total market sales attributed to the three most prominent sellers, as the measure of market concentration. CR3 is output at each time step.

$$\text{CR}_n = \frac{\sum_1^n SS_j}{\sum_1^{Ns} SS_j} \quad (5)$$

where $\sum_1^n SS_j$ signifies the cumulative sales of the top n sellers' sales, while $\sum_1^{Ns} SS_j$ represent the overall sales encompassing all sellers within the market.

4 Simulation experiments and results

A series of simulation experiments were conducted to investigate the influence of consumer characteristics and social interactions on market dynamics. Experiment 1 focused on examining the impact of consumer attention (CA) on online market concentration. Subsequently, robustness checks were performed to assess CA and market concentration relationships. The initial exploration in Experiment 2 involved investigating market size, using the number of sellers as a straightforward representation. Investigating market size is often valuable, as it directly influences competitive dynamics, consumer choice behavior, and market structure (Melitz and Ottaviano 2008). By understanding how the number of sellers affects market outcomes, we can gain insights into how variations in market size impact the overall market structure and the robustness of observed relationships between CA and market concentration. This can assist in determining whether the observed effects in smaller or larger markets are consistent and provide a comprehensive understanding of market dynamics under different conditions. Additionally, we manipulated the network connection coefficient (NCC) in Experiment 3 to measure the influence of peers on individuals. NCC was chosen as a key variable because it affects the extent and intensity of social interactions among individuals, which in turn impacts individual decision-making and market dynamics. By examining different levels of NCC, we can understand how the structure and strength of social networks influence individual decision-making, market dynamics, and the overall impact of social interactions on market outcomes. Notably, the intentional omission of product quality considerations in the experimental design aligns with the study's emphasis on markets featuring homogeneous products, particularly within the e-commerce domain, such as electronic goods with relatively uniform price levels. This choice is informed by the understanding that in such markets, consumer preferences and market dynamics are primarily shaped by factors—specifically, consumer attention, as addressed in this paper—rather than variations in product quality. The analysis underscores the coherence of marketing concepts and page layout in markets where standard products, exhibiting minimal variation in appearance, are prevalent.

The default settings of the model are as follows. First, the market consists of 200 consumers and 20 sellers who are randomly distributed. This sample size provides sufficient representation while considering model running time constraints. Second, the social connection coefficient (NCC) is set at 0.05, which, according to Eqs. 3 and 4, indicates that each consumer has an average of 5 link neighbors (4.975 to be precise). Third, we establish a default consumer attention level (CA) of 5, representing the initial intensity of attention consumers allocate to the market. It should be noted that the parameters in the model are adjustable and can be combined in various ways to suit different market environments. We conducted 20 replications of the experiments, and the reported results represent the average across these 20 replications. This approach is implemented to mitigate the inherent variability in the purchasing process, where consumers randomly select one option from the consideration set to make a purchase. The sales volume of the

chosen seller increases accordingly. The process then moves to the next round, repeating the same decision-making process. The model runs 200 steps, effectively capturing the unfolding dynamics of the market over this temporal span. The use of a 95% confidence interval in the presentation of results provides a statistical measure of the precision and reliability of the observed outcomes.

4.1 Experiment 1: the effect of limited attention

We first investigated whether and how changes in levels of consumer attention (CA), affect market concentration, specifically CR3. In Fig. 4, we present the dynamic trajectory of market concentration corresponding to low (CA = 2), medium (CA = 5), and high (CA = 8) values of consumer attention. The illustration reveals that diminished consumer attention corresponds to heightened market concentration. To elaborate, when CA is minimal (CA = 2), the aggregate market share of the top 3 sellers attains its zenith, reaching 60%. Conversely, at a heightened CA of 8, the concentration ratio experiences a reduction to approximately 12%. The concentration ratio (CR3) depicted in the figure demonstrates a pronounced increase, followed by a fluctuating state of balance.

This observed pattern is principally attributed to our methodology of accounting for new sales in each step, as opposed to cumulative sales. This approach allows for a more dynamic reflection of market activity, contrasting with cumulative sales that might inaccurately elevate sellers who have long been eliminated from the market still being at the top of the sales chart. Such unintended outcomes are effectively mitigated by our focus on new sales at each step.

To obtain a comprehensive understanding of the effects of consumer attention (CA) on concentration ratio (CR3), a series of experiments spanning CA

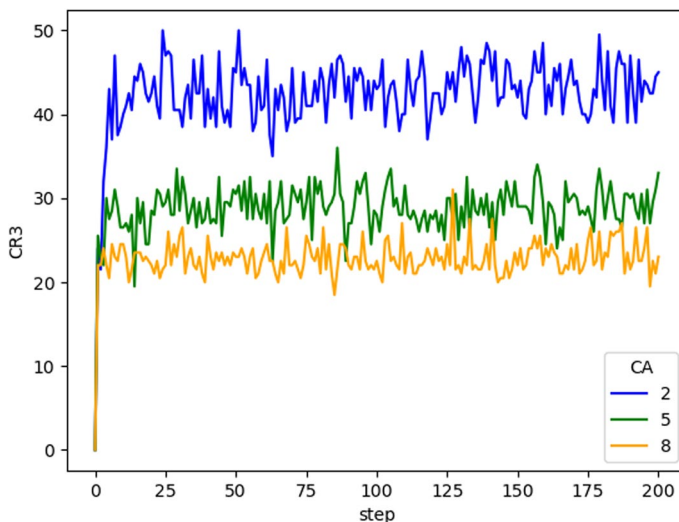


Fig. 4 Evolution of market concentration (CR3) under different levels of consumer attention (CA) in a typical run

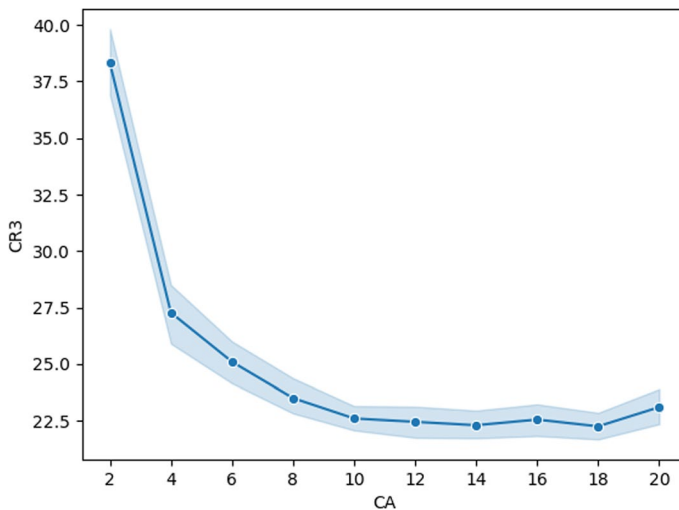


Fig. 5 Market concentration (CR3) with a 95% confidence interval under different levels of consumer attention (CA)

values ranging from 1 to 20 were undertaken. As illustrated in Fig. 5, a discernible inverse correlation between CA and CR3 emerges after 200 steps, a temporal threshold adequate for attaining market equilibrium. Notably, for CA values below 12, there is a dramatic decrease in CR3 from a high level, signifying a pronounced reduction in the adverse effects of CA on market concentration. In other words, the incremental impact of CA on CR3 diminishes.

Furthermore, for $CA > 12$, CR3 exhibited a slight fluctuation at a low level. This suggests that heightened consumer attention, indicative of increased random selection, leads to a concurrent reduction. In essence, a scarcity of consumer attention underlines the negative impact on market concentration; as attention increases, random selection leads to a decrease in market concentration.

This is because, according to the model, the products initially chosen by the local majority in the network of a consumer set the trend for subsequent consumers through social influence, leading to popular dominance even if the initial consumer choices are random. This social influence results in significant market inequality: popular products become more popular while unpopular products become less popular, irrespective of product quality. The choice influenced by social relations is always present in the consumer consideration set (the second item). When the CA is small, meaning there are fewer items to choose from in the consideration set, the second item is more likely to be selected, increasing the likelihood of consensus. Thus, the smaller the CA, the greater the effect of social influence, which more readily promotes market concentration. Conversely, when the CA is large, more randomly chosen products enter the consideration set, reducing the likelihood that socially influenced items will be selected, leading to a lower degree of market concentration.

4.2 Experiment 2: the sensitivity analysis of market size

In order to validate the robustness of our findings, specifically the inverse relationship between consumer attention and market concentration, we conducted a series of robustness checks. This experiment examined different market sizes, defined by the number of sellers (N_s). Initially, we assessed the influence of market size on market concentration independently, with the results illustrated in Fig. 6. Subsequently, we explored the combined effects of market size and consumer attention on market concentration, as shown in Fig. 7.

Figure 6 reveals a clear negative correlation between market size and market concentration, despite some variations. For instance, in smaller markets (e.g., $N_s = 10$), we observed an increase in market concentration, exceeding 40%. In contrast, larger markets (e.g., $N_s = 40$) demonstrated a significantly lower concentration ratio, falling below 20%. The negative relationship between market size and concentration ratio can be explained by the increased number of available products in larger markets. As market size grows, consumers are less likely to coordinate on the same products because of the greater number and the lack of significant differences among them. Therefore, the likelihood of any single product becoming dominant decreases, leading to lower concentration.

Figure 7 depicts a more complex three-way interaction effect between consumer attention (CA), market size (N_s) and market concentration (CR3). In general, the graph's color transitions from bright yellow to dark brown from left to right and bottom to top, indicating a gradual decrease in CR3 with an increase in either the number of CAs or N_s . This pattern aligns with our previous experimental findings, demonstrating that consumer attention has a negative impact on market concentration, even when the market size varies.

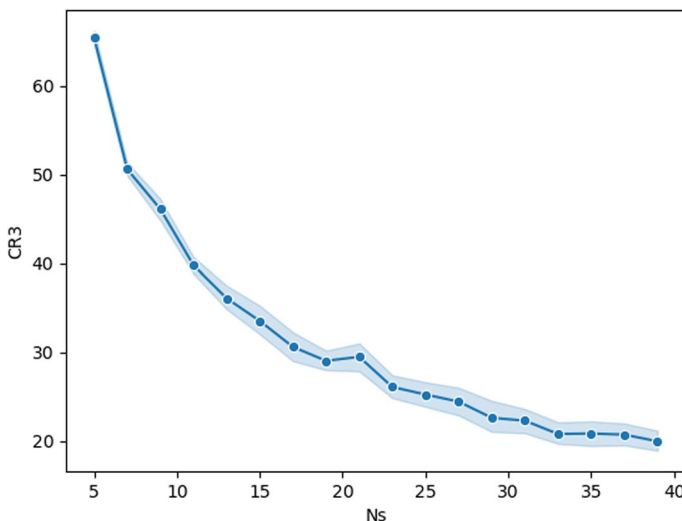


Fig. 6 Market concentration (CR3) with 95% confidence interval under different levels of market size (N_s)

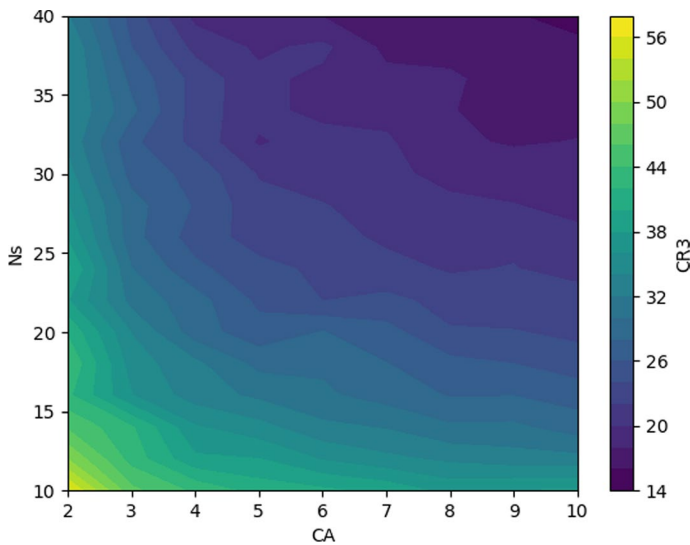


Fig. 7 Contour plot of market concentration (CR3) with the interactions of consumer attention (CA) and market size (N_s). The color represents the degree of market concentration, where blue indicates the least concentration and yellow the most

4.3 Experiment 3: the sensitivity analysis of social connections

In this experiment, we further investigated the sensitivity to the network connection coefficient (NCC). As illustrated in Fig. 8, we explore the correlation between market concentration and NCC. In general, CR3 is not significantly affected by variations in NCC. It is observed that CR3 shows an alternating trend of downward and upward fluctuations as NCC increases. Although a low point of CR3 is observed at the intermediate NCC value of around 0.6 and a peak of CR3 can be seen at the maximum and minimum values, the difference is not significant enough.

This result seems somewhat counterintuitive. Intuitively, it might be expected that the higher the network connection density, the more likely it is for a consensus decision to be reached quickly, leading to higher market concentration. However, this outcome did not emerge in this model. The reason is that as the NCC increases, the network resembles a more random structure, exposing individuals to many diverse "peer influences." This diversity makes coordination more challenging. At the same time, the average path distance decreases, allowing information to travel faster. The interplay of these two effects could explain the observed nonlinear pattern.

Figure 9 displays the collective impact of NCC and CA on CR3. The colors in the figure gradually darken from left to right, demonstrating that CR3 decreases steadily as CA increases, which is in agreement with the findings of Experiment 1. We also see that the dark blue clusters are distributed at multiple positions in the middle of the figure, illustrating that the interaction between NCC and CA produces a complex outcome.

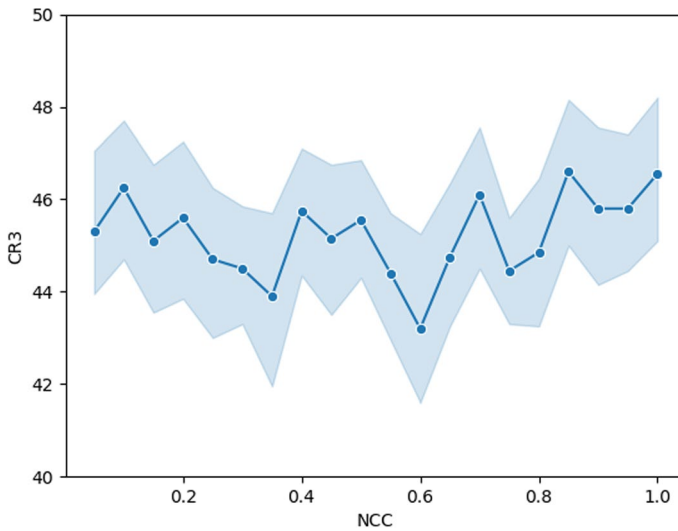


Fig. 8 Market concentration (CR3) with 95% confidence interval under different levels of network connection coefficient (NCC)

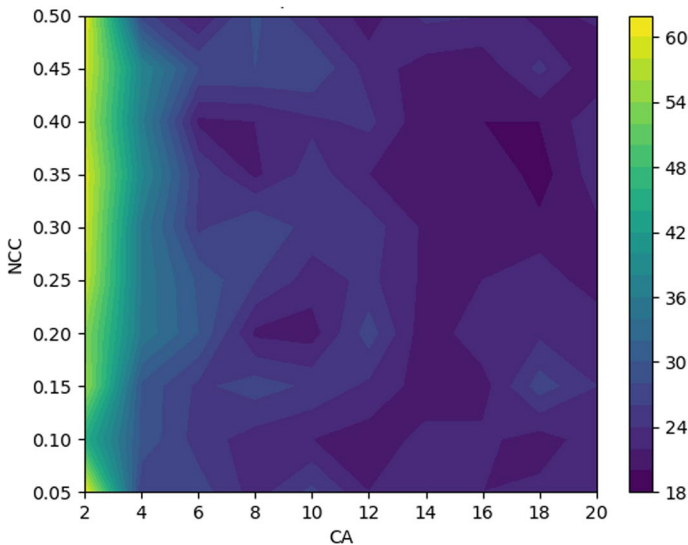


Fig. 9 Contour plot of market concentration (CR3) with the interactions of consumer attention (CA) and social connection coefficient (NCC). The color represents the degree of market concentration, where blue indicates the least concentration and yellow the most

The current experiments are mainly to demonstrate our model's ability to experiment with consumer interactions and identify their impact on market concentration. Future experiments will be required to gain a more comprehensive understanding of

the interplay between consumer attention and market dynamics. Additionally, further sensitivity analyses could enhance the robustness of the findings. The experiments presented here serve as a proof of concept, illustrating the feasibility of conducting diverse simulations.

5 Discussion

The observed negative correlation between consumer attention and market concentration reveals compelling insights into consumer behavior and market dynamics. In a hypothetical scenario where sellers and their products are indistinguishably identical, rational consumers would be expected to make random choices. Consequently, the market should ideally exhibit a state of perfect competition, with all sellers having an equal market share. However, the experimental findings significantly deviated from this theoretical expectation, revealing a notably high concentration. This discrepancy is primarily due to the introduction of a more realistic assumption in the model, specifically the constraint of consumer attention and social influence. The more limited consumers' attention is, the more likely they are to adopt heuristic decision-making patterns (Hafenbrädl et al. 2016), making them susceptible to the influence of social norms and their past purchasing experiences. As a result, products that are initially chosen randomly by the majority can maintain a sales advantage, leading to higher market concentration. This is similar to the concept of social lock-in, as discussed by Nyborg et al. (2016), where societal norms and behaviors become entrenched or "locked in," making it challenging to transition toward more sustainable or socially desirable behaviors despite the presence of alternatives. However, it is worth mentioning that the market-leading sellers, in this scenario, do not attain their position through product dominance, and their leading position is not necessarily fixed. In light of these dynamics, the concentration in this market can be referred to as "pseudo-concentration," making it apart from the conventional market economy, where dominance is typically determined by product attributes and market forces. This negative relationship between consumer attention and market concentration persists across variations in market size and consumer network connections.

The assumption of product uniformity in the model is not intended to dismiss the relevance of price or quality, but rather to control for variables and focus specifically on the impact of consumer behavior, particularly interactions through networks, on market concentration. Consumers engage with a variety of multiple information channels, and price or quality is only one of them, often not the most salient one in their decision-making, especially online, where many products with similar prices can be quickly compared. While it seems intuitive that market concentration would result from all consumers choosing the lowest-priced or highest-quality products, this does not fully reflect reality. Consumers are influenced by social dynamics—such as word of mouth (WoM) in our model—which affects their choices in ways that go beyond pure economic rationality. The assumption of product homogeneity in our model allows for a controlled focus on understanding how consumer behavior, rather than product differentiation, contributes to market concentration.

We focus on the e-commerce market to illustrate how micro-level behaviors, particularly those of consumers, can lead to unexpected macro-level phenomena. This is because, firstly, the information overload inherent in e-commerce environments amplifies the role of limited consumer attention, making it a critical factor in decision-making. Secondly, e-commerce provides a unique setting for studying the dynamics of social influence and word of mouth, as online platforms facilitate rapid dissemination of information and peer reviews. Importantly, the insights and mechanisms from our model can be adapted to offline markets, where factors such as physical proximity and local social networks play analogous roles to online reviews and digital connectivity. Notably, as the model is inherently theoretical, it is also adaptable to other similar social contexts where bounded rationality and social influence drive decision-making processes.

Previous literature has predominantly analyzed seller-side factors to explain market concentration. However, our research shifts the focus to examining how consumer interactions, particularly in contexts with homogeneous products (such as in the e-commerce realm), influence market concentration. Other relevant theoretical models for explaining consumer behavior are CONSUMAT I (Jager 2000), CONSUMAT II (Jager and Janssen 2012), and the recent HUMAT version (Antosz et al. 2019). These models are well-established for simulating behavioral processes and explaining how psychological and social factors affect market dynamics. It primarily bases consumer motivations on satisfaction and uncertainty, aiming to create a meta-theoretical or integrative framework that synthesizes various choice theories (Brouwers and Verhagen 2003). While this study draws initial inspiration from CONSUMAT, it simplifies and adapts the model to fit its specific context. Rather than adopting CONSUMAT directly, we construct an independent framework that shares key assumptions with CONSUMAT, such as the need for cognitive and time resources in decision-making, represented here by “limited attention.” Additionally, both models recognize social influences in consumer decision-making, reflected in our model through the variable “word of mouth” (WoM).

The models developed in this paper provide valuable insights for understanding how the concentration pattern at the market level can emerge from the interactions at the consumer level. However, this study is not without limitations. Several issues need to be addressed in future work to enhance the robustness of our models. First, the behaviors and interactions of agents in agent-based models often serve as simplified representations of reality. Such simplification is a strategic choice to facilitate a more focused understanding of specific mechanisms. However, it may not capture the full complexity of consumer decision-making processes. For example, our model simplifies consumer decision-making into three stages: search, inquiry, and purchase, each governed by a singular behavioral rule. This is not an exactly truthful description of the real-world shopping process. Future research can add more social complexity by considering additional consumer stages and enriching behavioral mechanisms to enhance the model’s explanatory power. Second, there is room for refinement in defining and quantifying limited attention. Future research could delve into developing more sophisticated measures and models that capture the multifaceted nature of attention, considering factors like selective exposure, information processing capacity, and the temporal aspects of attention allocation.

Validation is a key challenge in developing agent-based models. This difficulty arises from the dynamic and emergent nature of the simulated system. Validating ABMs is possible in principle (Fagiolo et al. 2019; Moss and Edmonds 2005; Windrum et al. 2007). For instance, survey data can be used to measure how much attention consumers allocate to various products, and these responses can inform the assignment of weights to different information channels. Experimental techniques, such as eye-tracking, can also quantitatively measure the time consumers spend on specific elements, allowing for precise calibration of attention in the model. In our study, we did not calibrate model parameters in the strict technical sense of systematically adjusting parameters to align the model with observed data; rather, parameters are based on plausible assumptions informed by existing literature or theoretical perspectives. The ABM models developed here have revealed several potential mechanisms of consumer decision-making and their macroscopic consequences of market concentration. Due to their theoretical and abstract nature, these mechanisms cannot be directly mapped into reality. Nevertheless, they may assist researchers in identifying relevant stylized facts when observing social phenomena, and in understanding the underlying social processes behind these facts. At this stage, our models serve as a computational theoretical experiment that allows us to simulate and examine the potential outcomes of different assumptions and scenarios. The validity of these assumptions and the resulting conclusions hinges on future empirical verification.

6 Conclusions

Agent-based modeling emphasizes the heterogeneity, bounded rationality and learning of agents. We have constructed a simulation model that utilizes consumer interaction and learning mechanisms to reveal alternative patterns of market concentration. In this model, consumers are bounded rational in terms that they can only exercise limited searches each time. Newly input information and knowledge will affect consumers' evaluation of sellers and, in turn, their decision pools. When buying again, consumers will compare previous selections and neighbor recommendations, and the purchase will be made with the most trusted sellers. The model illustrates that consumers, through their choices among essentially identical brands, can generate market concentration based on subjective or intersubjective preferences.

This research underscores the critical role of consumers in shaping market concentration, providing a complementary perspective to existing market structure studies and potentially introducing new aspects into traditional industrial organization theory. In the context of the real market economy, where information overload and product homogeneity are prevalent, the importance of attention economics becomes increasingly evident. It is worth noting that many search engines and recommendation algorithms employed by e-commerce platforms may accelerate the consumer decision-making process. However, these algorithms can only influence which products enter a consumer's consideration set; they cannot change the size of the consideration set itself. Put another way, they have a built-in "rich get richer" dynamic and do not fundamentally address the issue of limited consumer

attention. Therefore, being memorable becomes crucial for sellers in intense competition. This is particularly relevant in online markets, where traditional theories struggle to explain concentration phenomena.

As a theoretical modeling paper, our main goal is to enhance comprehension of specific social phenomena rather than proposing specific policy recommendations. For example, our research can help shed light on the phenomenon of “E-commerce + Internet Celebrity.” Internet celebrities themselves serve as a means of capturing attention and creating focal points, with their value increasing as attention becomes scarce. This is one of the reasons for the enormous profits of Internet celebrities in the online economy. However, our simulation results can provide practical implications for various market stakeholders, such as marketers, e-commerce platform designers, and policymakers.

For marketers, in light of restricted consumer attention, product marketing strategies should prioritize attracting and retaining consumers’ interest and attention. They can develop more targeted advertising strategies that resonate with targeted consumers amidst information overload. With attention becoming a scarce resource in competitive markets, marketers need to focus on creating memorable brand experiences that differentiate their products from competitors, which involves emphasizing unique selling propositions, storytelling, and emotional branding. For E-commerce platforms, they may try to enhance user experience by implementing features that facilitate consumer attention and engagement. Designers can optimize content curation and presentation to highlight featured products, promotions, and customer reviews, guiding users toward items of interest and encouraging exploration within the platform. Similarly, policymakers can use the theoretical insights provided to examine the broader implications of consumer-driven market concentration. While this study does not simulate specific e-commerce platforms, it suggests that future policies consider the significant role of consumer-side factors in shaping market outcomes.

Model document: The model was implemented with NetLogo 6.0.4. The full code and model are openly available on the CoMSES Computational Model Library at: <https://www.comses.net/codebase-release/03235203-3fc2-453b-86dc-f2325b0d8190/>.

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Author contributions Conceptualization was performed by Teng Li, Shaoni Wang, and Deyi Zhou; data curation by Teng Li and Shaoni Wang; formal analysis by Teng Li and Amar Razzaq; investigation by Teng Li, Shaoni Wang, and Deyi Zhou; methodology by Teng Li and Shaoni Wang; project administration by Deyi Zhou; resources by Deyi Zhou; software by Teng Li and Shaoni Wang; supervision by Deyi Zhou; validation by Amar Razzaq; visualization by Teng Li; writing—review and editing—by Amar Razzaq.

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Declarations

Conflict of interest The authors declare no conflict of interest.

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